

IMPLEMENTING LOGGING WITH SPRING AOP (ASPECT ORIENTED PROGRAMMING)

pom.xml (Updated with Spring AOP):

```
<dependencies>
  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-context</artifactId>
    <version>6.1.14</version>
  </dependency>

  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-aop</artifactId>
    <version>6.1.14</version>
  </dependency>

  <dependency>
    <groupId>org.springframework</groupId>
    <artifactId>spring-webmvc</artifactId>
    <version>6.1.14</version>
  </dependency>

  <dependency>
    <groupId>org.aspectj</groupId>
    <artifactId>aspectjweaver</artifactId>
    <version>1.9.22.1</version>
  </dependency>
</dependencies>
```

LoggingAspect.java (New):

```
package com.library.aspect;

import org.aspectj.lang.ProceedingJoinPoint;
import org.aspectj.lang.annotation.Around;
```

```

public class LoggingAspect {
    @Around("execution(* com.library.service.BookService.*(..))")
    public Object logExecutionTime(ProceedingJoinPoint joinPoint) throws Throwable {
        long start = System.currentTimeMillis();
        Object result = joinPoint.proceed();
        long end = System.currentTimeMillis();
        System.out.println(joinPoint.getSignature().getName()
            + " executed in " + (end - start) + " ms");
        return result;
    }
}

```

applicationContext.xml (Updated):

```

<?xml version="1.0" encoding="UTF-8"?>

<beans xmlns="http://www.springframework.org/schema/beans"
    xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xmlns:aop="http://www.springframework.org/schema/aop"
    xsi:schemaLocation="
        http://www.springframework.org/schema/beans
        https://www.springframework.org/schema/beans/spring-beans.xsd
        http://www.springframework.org/schema/aop
        https://www.springframework.org/schema/aop/spring-aop.xsd">

    <bean id="bookRepository"
        class="com.library.repository.BookRepository"/>

    <bean id="bookService"
        class="com.library.service.BookService">

        <property name="bookRepository" ref="bookRepository"/>

    </bean>

    <bean id="loggingAspect"
        class="com.library.aspect.LoggingAspect"/>

    <aop:aspectj-autoproxy/>
    <aop:config>

```

```

<aop:aspect ref="loggingAspect">
    <aop:around method="logExecutionTime"
        pointcut="execution(* com.library.service.BookService.*(..))"/>
    </aop:aspect>
</aop:config>
</beans>

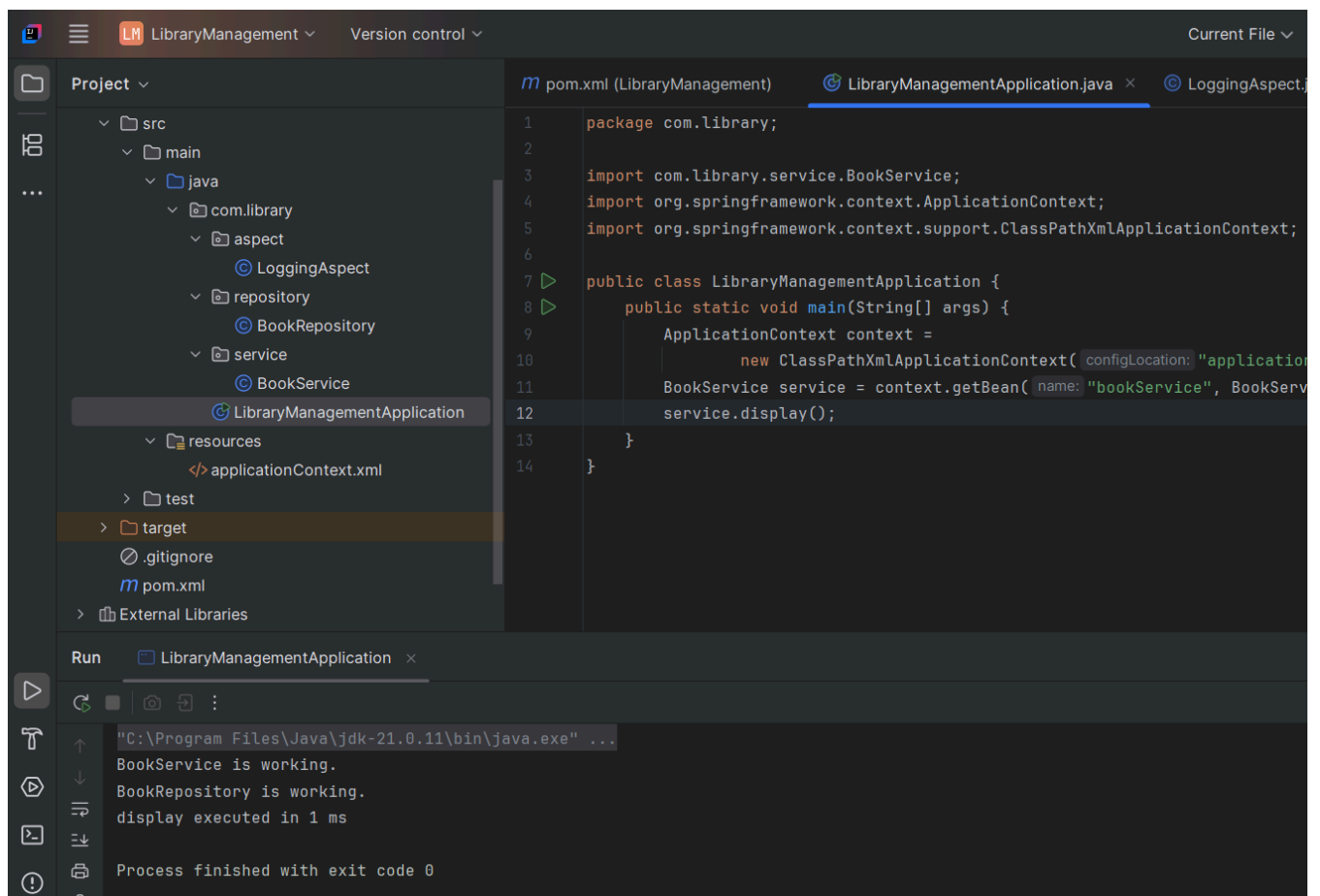
```

LibraryManagementApplication.java and other files:

No changes, same as the previous exercise

Output:

“display executed in 1 ms”:



“display executed in 2ms”:

